

SIMATS ENGINEERING

ASSESSMENT TOOL 2: Industry Problem Solving Task

COURSE NAME: Cloud Computing and Big Data Analytics **COURSE CODE:** CSA15

COURSE OUTCOME COVERED

CO4: *Analyze big data characteristics and challenges across domains including security and application aspects.*
(BL4)

Dave's Taxonomy: Articulation (Level 4)
Question Count: 5

Total Marks: 40

Duration :60 Minutes

Code of Conduct

I _____ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: _____

Assessment Tasks

Industry Problem Solving Task

1. **Retail Data Optimization Problem** - A retail chain operates both physical stores and an online platform. It generates large volumes of sales, inventory, and customer behavior data. The company wants real-time inventory tracking and demand forecasting. It also aims to deliver personalized marketing recommendations to customers. Design a big data architecture to support these requirements. Explain the tools, technologies, and analytics models you would use.
2. **Social Media Fraud Detection Problem** - A social media platform processes billions of user interactions daily. These include posts, comments, likes, and shares. The platform wants to detect fake accounts and misinformation quickly. Real-time or near real-time processing is required. Design a scalable big data solution for this use case. Explain the role of machine learning and stream processing frameworks.
3. **Government Data Platform Problem** A government agency is building a national data platform. It combines census, economic, and geospatial datasets. Different departments need access to shared and integrated data. The system must ensure interoperability across multiple agencies. Strict privacy and security requirements must be followed. Propose a governance and data management strategy.
4. **Banking Fraud Detection Problem**
A bank processes millions of financial transactions every minute. It needs to detect fraudulent activities in real time. The system must analyze historical and streaming transaction data. Low latency and high accuracy are critical requirements. Design a big data pipeline for fraud detection. Explain the technologies and algorithms you would implement.

5. Agriculture Smart Farming Problem

A smart farming system collects soil, weather, and crop data. Sensors and drones provide real-time field information. Farmers want insights to improve yield and resource usage. Data is generated from multiple distributed sources. Design a big data solution for precision agriculture. Explain how analytics can support decision-making.

Industry Problem Solving Rubric

Criteria	Weightage	Level 4 - Exemplary	Level 3 - Proficient	Level 2 - Developing	Level 1 - Beginning
Understanding of Industry Context	20%	Demonstrates strong understanding of real-world problem context	Good understanding	Basic understanding	Weak understanding
Application of Big Data Concepts	25%	Applies highly relevant big data ideas and tools	Mostly appropriate application	Partial application	Weak application
Solution Design	25%	Solution is practical, scalable, and well-reasoned	Reasonable solution	Basic solution with gaps	Unclear solution
Security / Risk Awareness	15%	Strong awareness of security and operational risks	Reasonable awareness	Limited awareness	Poor awareness
Clarity and Professionalism	15%	Well-structured and professional response	Generally clear	Moderately clear	Poorly presented

Total Marks Awarded:

1. **Retail Data Optimization Problem** - A retail chain operates both physical stores and an online platform. It generates large volumes of sales, inventory, and customer behavior data. The company wants real-time inventory tracking and demand forecasting. It also aims to deliver personalized marketing recommendations to customers. Design a big data architecture to support these requirements. Explain the tools, technologies, and analytics models you would use.
 - The platform handles billions of website clicks, mobile interactions, and customer transactions. It would likely use HDFS or cloud storage such as Amazon S3 for distributed storage. Apache Kafka can be used to collect and transport data from websites, mobile applications, and transaction systems.
 - Structured data such as customer and transaction records can be stored in relational databases, while semi-structured data such as clickstream logs can be stored in JSON, Avro, or Parquet format. Data can be partitioned based on date, region, or event type to improve query performance.
 - Apache Spark can be used for large-scale batch processing, while Spark Streaming or Apache Flink can process real-time events. Replication and horizontal scaling help the system handle failures and increasing workloads.
 - Finally, processed data can be stored in a data warehouse and connected to Power BI or Tableau for dashboards showing sales, customer behavior, and other business KPIs.
2. **Social Media Fraud Detection Problem** - A social media platform processes billions of user interactions daily. These include posts, comments, likes, and shares. The platform wants to detect fake accounts and misinformation quickly. Real-time or near real-time processing is required. Design a scalable big data solution for this use case. Explain the role of machine learning and stream processing frameworks.
 - An e-commerce platform generates data from product views, cart additions, abandoned checkouts, and payments. Apache Kafka would likely be used to stream these events in real time. Customer profiles can be stored using distributed databases such as MongoDB, Cassandra, or HBase.
 - Redis can be used as a caching system to provide fast access to frequently used customer and product information. Clickstream, session, and transaction data can be synchronized using customer ID, session ID, and timestamps.
 - Apache Spark or Flink can process the data, while Spark SQL, Trino, or Presto can perform distributed queries. Machine learning models can analyze customer behavior and generate personalized product recommendations.
 - Structured order data can be combined with semi-structured browsing data for customer analytics. Kafka replication, database replication, and partitioning provide fault tolerance and scalability. Power BI or Tableau can display KPIs such as sales, conversion rate, and cart abandonment.
3. **Government Data Platform Problem** - A government agency is building a national data platform. It combines census, economic, and geospatial datasets. Different departments need access to shared and

integrated data. The system must ensure interoperability across multiple agencies. Strict privacy and security requirements must be followed. Propose a governance and data management strategy.

- The healthcare system manages patient histories, medical images, wearable sensor data, and prescription records. A hybrid database architecture would be suitable. Relational databases can store structured information such as patient records and prescriptions, while NoSQL databases and cloud storage can store documents, sensor data, and medical images.
- HL7 and FHIR standards can be used to exchange healthcare information between hospitals, laboratories, and insurance systems. Apache Kafka can collect continuous wearable-device data, while Spark Streaming or Flink can process it in real time.
- Machine learning models can be developed using historical patient data for disease prediction, risk assessment, and patient monitoring. Apache Spark can perform large-scale population health analysis.
- Because healthcare information is sensitive, the system should use encryption, authentication, role-based access control, and audit logs. Analytics dashboards can provide doctors and researchers with patient trends, risk scores, and population-level insights.

4. **Banking Fraud Detection Problem** - A bank processes millions of financial transactions every minute. It needs to detect fraudulent activities in real time. The system must analyze historical and streaming transaction data. Low latency and high accuracy are critical requirements. Design a big data pipeline for fraud detection. Explain the technologies and algorithms you would implement.

- A smart city platform collects data from traffic sensors, CCTV systems, weather stations, and public transport. Edge computing can process some data close to the sensors before sending it to the central system. This reduces network traffic and improves response time.
- Apache Kafka or MQTT can be used for continuous data transmission. Historical information can be stored in HDFS or cloud storage. Time-series databases such as InfluxDB or TimescaleDB can store sensor readings, while PostGIS can support geographical and location-based queries.
- Spark Streaming or Apache Flink can analyze traffic data in real time and predict congestion. Batch processing can analyze historical traffic patterns for urban planning.
- Security can be provided through encryption, authentication, role-based access control, and audit logging. Tools such as Power BI, Tableau, or Grafana can provide live dashboards showing traffic, weather, transport activity, and incidents.

5. **Agriculture Smart Farming Problem** - A smart farming system collects soil, weather, and crop data. Sensors and drones provide real-time field information. Farmers want insights to improve yield and resource usage. Data is generated from multiple distributed sources. Design a big data solution for precision agriculture. Explain how analytics can support decision-making.

- This system combines patient records, wearable readings, laboratory reports, and appointment histories. A hybrid storage architecture can use relational databases for structured medical records, NoSQL databases for flexible data, and a secure data lake for large-scale storage.
- ETL pipelines can extract data from different hospitals and clinics, transform it into a common format, and load it into the analytical system. HL7 and FHIR can help different healthcare systems exchange information.

- Wearable-device data can be processed in real time using Kafka, Spark Streaming, or Flink. Apache Spark can process millions of patient records for population-level healthcare analysis.
- Machine learning models can be used for disease-risk prediction, diagnosis support, patient deterioration prediction, and treatment recommendations. Healthcare data must be protected using encryption, authentication, access control, and audit mechanisms. Dashboards can provide doctors with patient trends and risk scores to support clinical decisions.